

**BitFlip**

# **CloudSherpa**

## **User Manual**

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# 1 | Welcome to CloudSherpa

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CloudSherpa is an intelligent cloud cost management platform designed to help organizations gain complete visibility into their cloud spending across multiple cloud providers. By combining cloud usage, costs and insights into a single, intuitive finOps dashboard, it eliminates the need to navigate between multiple cloud provider platforms.

The platform enables users to monitor cloud expenditure, identify unexpected cost increases and receive proactive recommendations to optimize resource usage and reduce unnecessary spending. Through real-time insights and predictive analytics, CloudSherpa helps organizations make informed financial decisions while maintaining efficient cloud operations.

By providing a unified view of multi-cloud environments and actionable cost optimization recommendations, CloudSherpa empowers businesses to control cloud expenses, maximise the values of their cloud investments and confidently manage their cloud platform.

## Do I need to be a technical expert?

CloudSherpa is designed for both technical and non-technical users. It features a user-friendly dashboard that presents insights in a clear, intuitive manner, which enables users of any technical level to easily understand what is occurring.

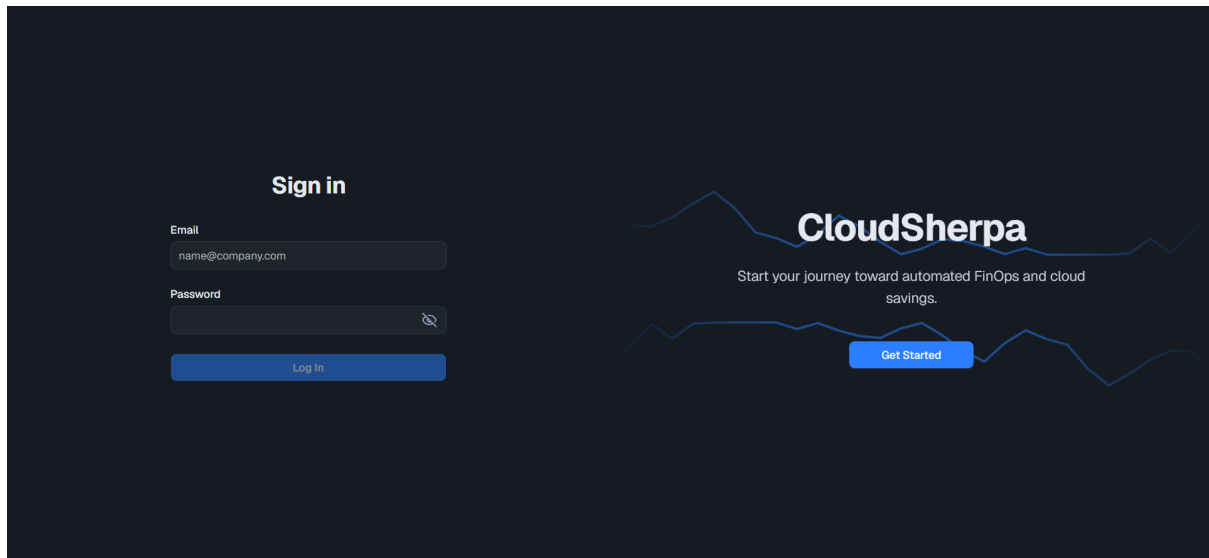
## 2 | Getting Started

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### 2.1 Accessing the System and Logging In

How to access the live deployment of CloudSherpa:

1. Navigate to <https://cloudsherpa.gjjcs.org/>



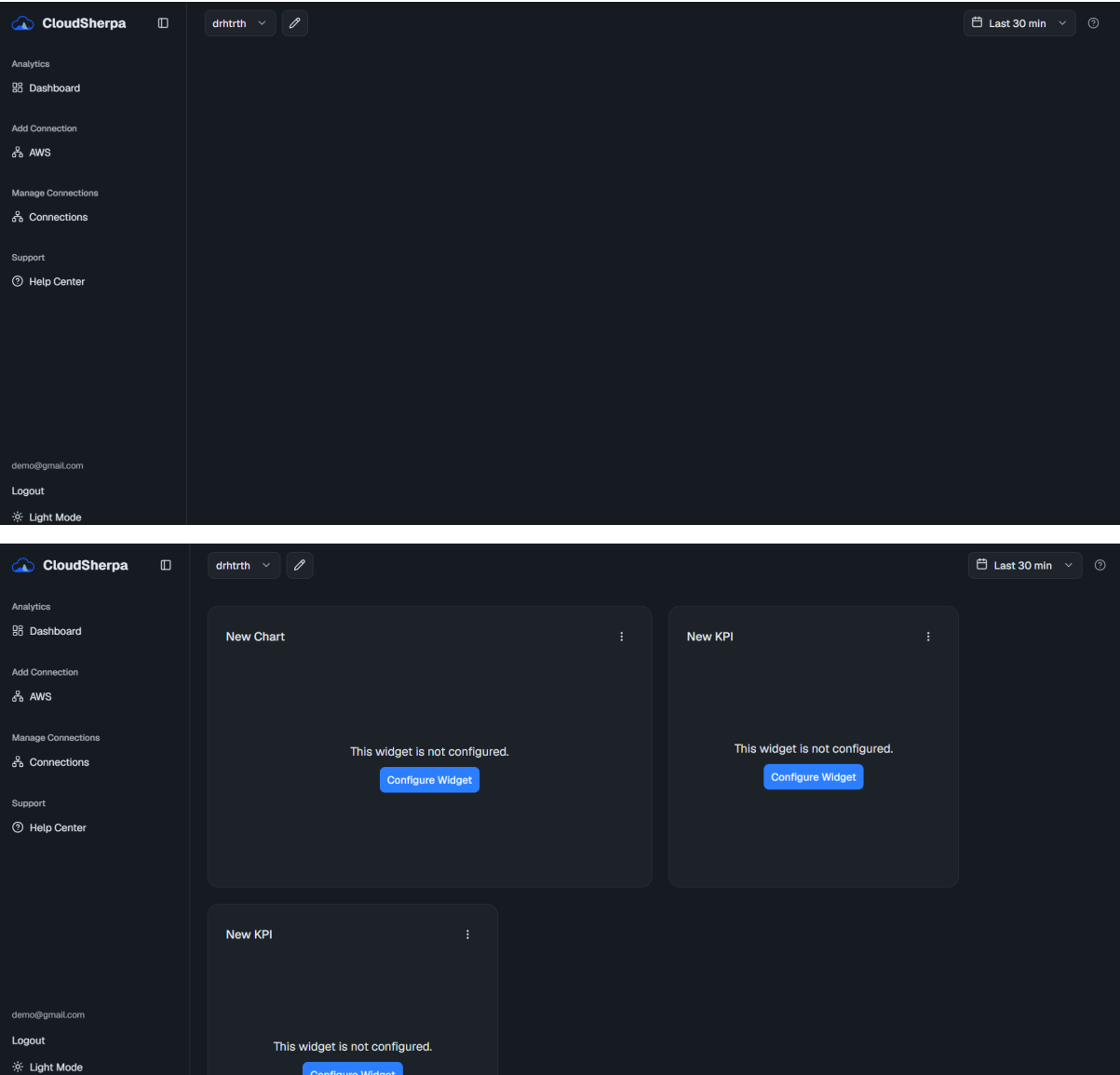
To create an account with CloudSherpa, follow these steps:

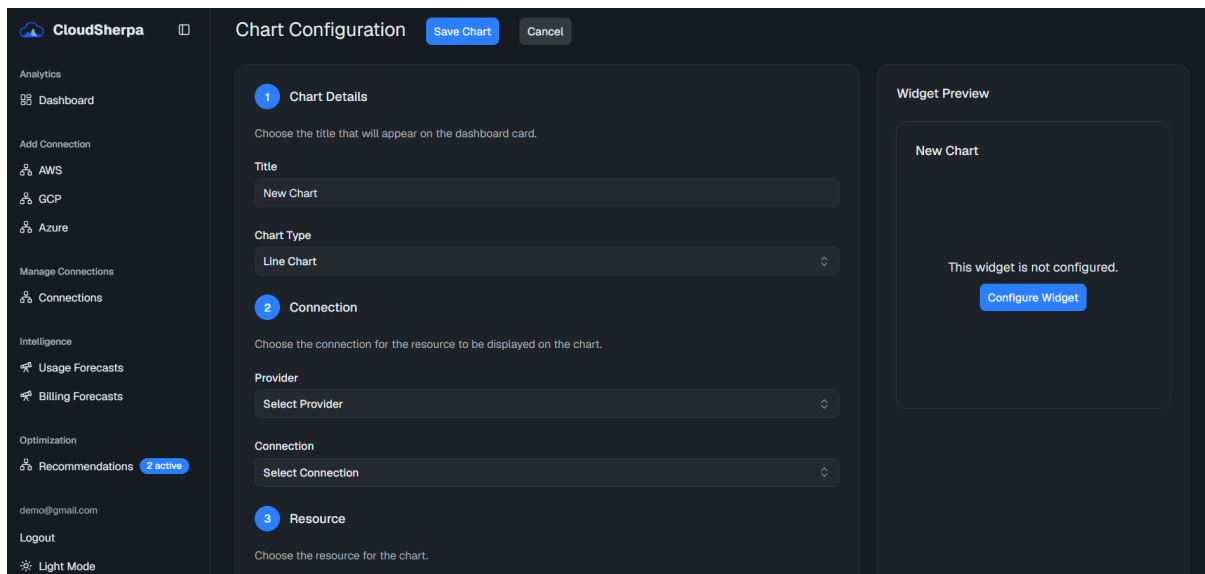
1. If you are a new user:
  - (a) Click the **Register** button.
  - (b) Fill in the required information.
  - (c) Create a secure password and confirm your registration.
2. If you already have an account:
  - (a) Click the **Login** button.
  - (b) Enter your registered email and password.
  - (c) Click **Login** to access your dashboard.

By following these steps, you can immediately begin using CloudSherpas features.

## 2.2 The Main Dashboard

Upon logging in the user is directed to the dashboard:





Key areas of the dashboard:

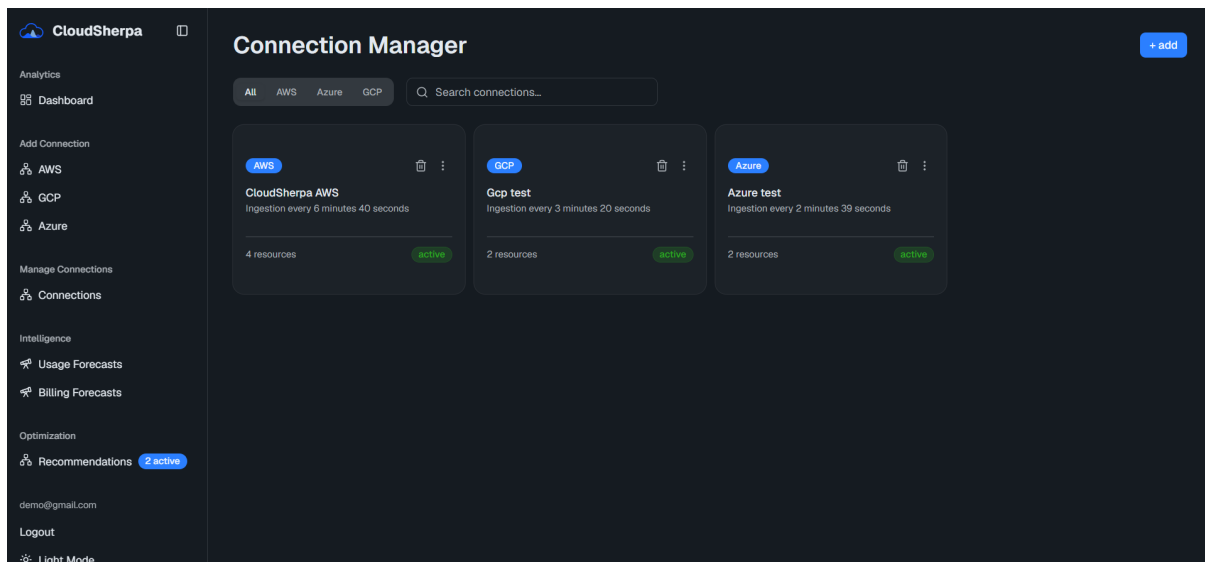
- **Navigation bar:** Users are able to navigate to various connections such as AWS. They are able to log out via the navigation bar or go back to the dashboard. The navigation bar is collapsible to give users a better view of what is being displayed on their screen.
- **Period selector:** The user can select which time period they would like to view the data being displayed. There are fixed period selectors such as 'last 7 days' or they can create a custom period that they desire.
- **Multiple dashboard:** More than one dashboard can be created via the dropdown. The users can also navigate to dashboards that were previously created via the dropdown.
- **Customize widgets:** Users can design the dashboard in a way they desire. The widgets size can be adjusted and their positions can be changed. This can be done by using the pen icon (next to the dropdown). They can further choose a graph they would like the data to be displayed in, from the options provided. The name of the widget can be customized and different types of data can be selected to be displayed eg.CPU.

## 3 | Core Features and Usage

These are the five core features of CloudSherpa

### 3.1 Feature 1: Connection Manager

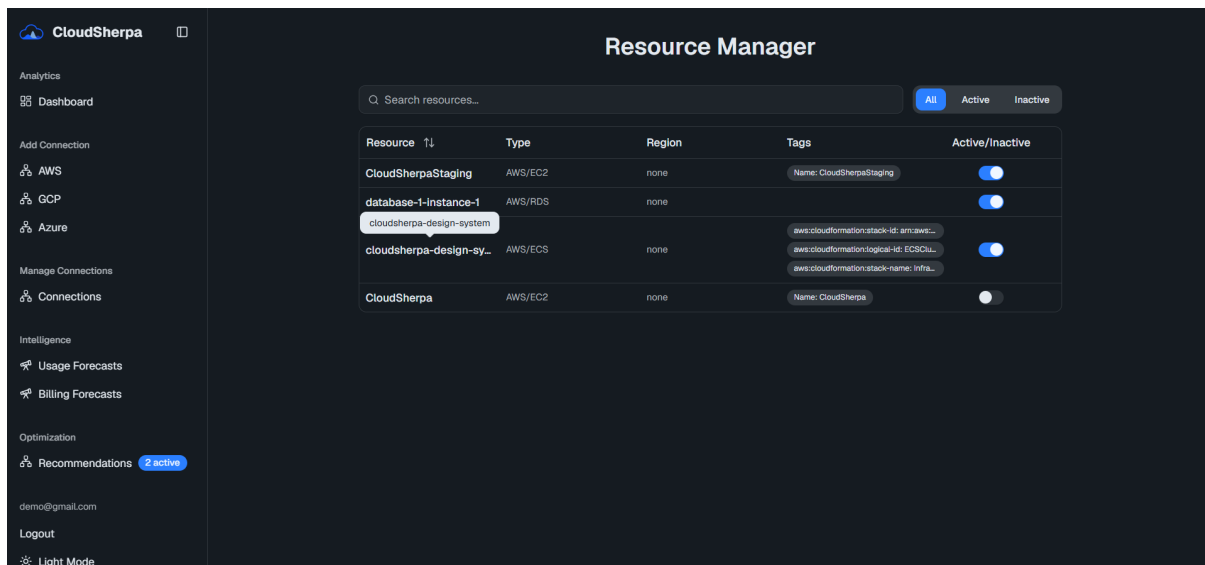
The connection manager is utilized to add multiple AWS connections where their services can be monitored as well as specific resources.



1. Locating connection manager - Navigate to the Connection manager (under Manage Connections) in the side navigation bar
2. For adding a connection you can navigate to the connection manager and click on the add button or you can use the side navigation bar to choose which connection you would like to establish.
3. Managing your connections - by clicking on the view details option. (will get by clicking on the 3 dots) You can change the name of the connections and view the resources that is associated with that connection. Multiple connections can be added and deleted.

## 3.2 Feature 2: Resource Manager

The resource manager is used to manage your connections. You can make certain resources active and inactive. You can filter and search for a resource.



1. Locating the resource manager - In the connection manager configuration (was shown in previous feature), click on the edit resources button. You can successfully configure your resources here.
2. Active/Inactive resources - toggle the resources to determine which resource should be monitored for that particular service or not.
3. Search and filter resources - Use the search bar to search for a resource or to filter click on the Resource name.
4. tooltips have also been included for resource names, types and tags hence the full view can be viewed when hovered over them.

## 3.3 Feature 3: Scheduler

The scheduler scans the database every 30 seconds for the usage scheduler and 5 minutes for the billing scheduler.

We read in a field that informs us that an account is due and if they are they get enqueued to Jobrunr background jobs queue which is managed by Jobrunr.

Jobrunr spins threads in the background, up to 8 in our configuration. Each job attempts ingestion for only one account. If it fails, it retries 3 times and if it has still failed



the appropriate field in the database will not get updated which mean that when the scheduler scans again that account will be scheduled again.

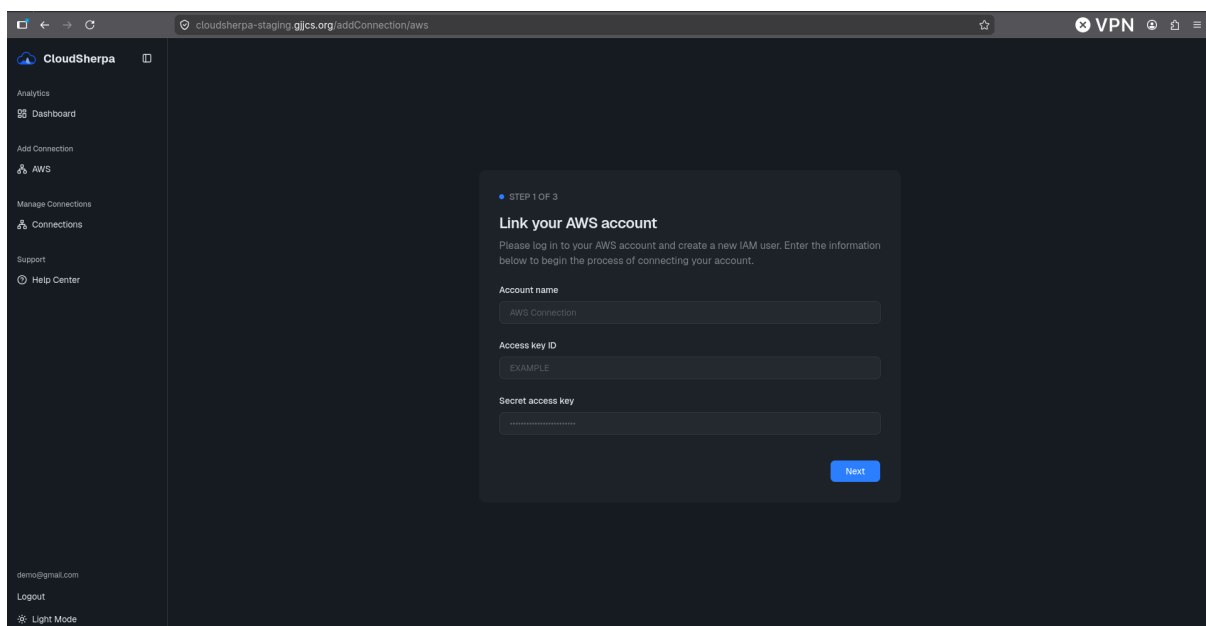
If it was successful, the appropriate next ingestion field gets updated in the database such that the next scan will not pick the same job again

## 3.4 Feature 4: Setup wizard

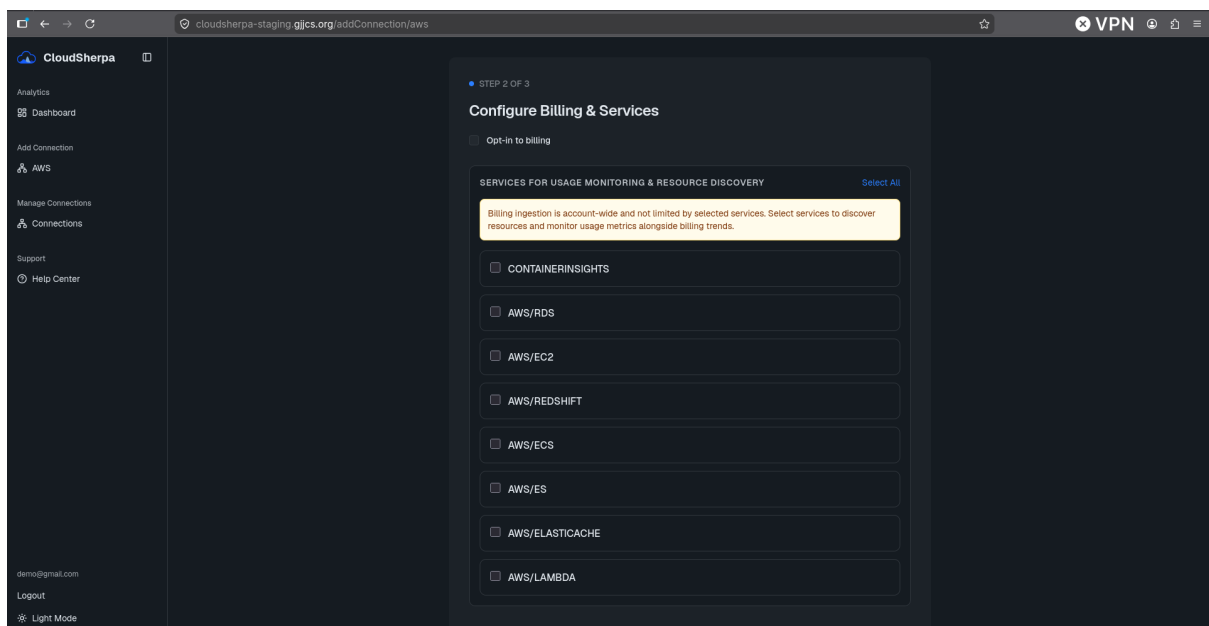
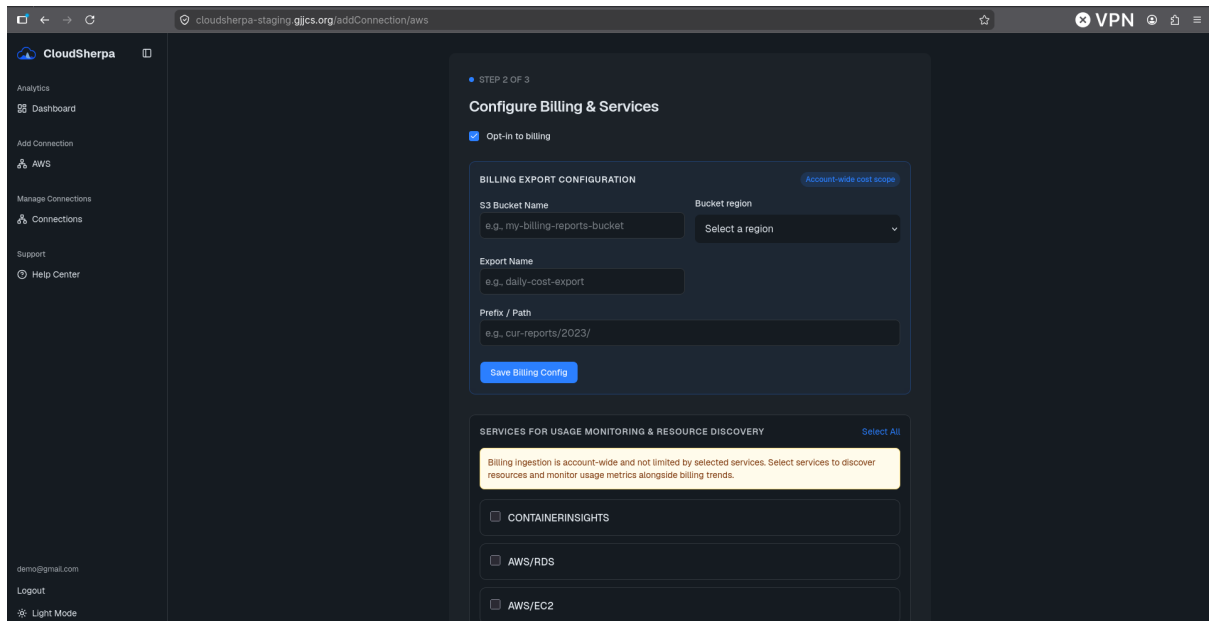
CloudSherpa provides three setup wizards for connecting to different cloud providers:

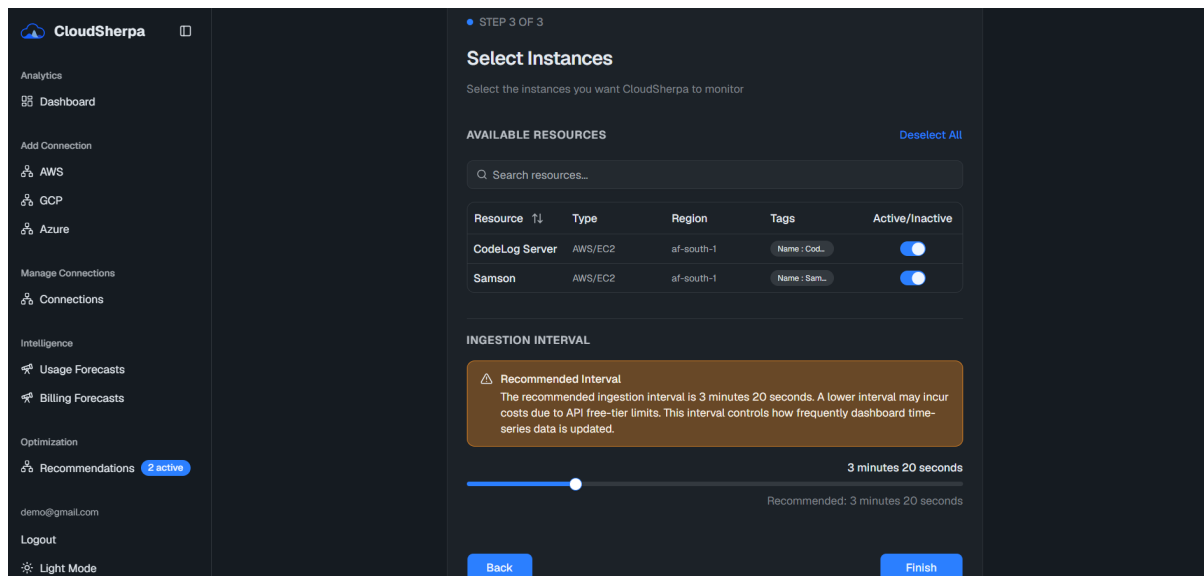
### a) AWS Setup Wizard

The setup wizard consists of 3 steps, which each user has to go through each time they want to establish an AWS connection.



The screenshot shows the CloudSherpa web application interface. On the left is a dark sidebar with navigation links: Analytics, Dashboard, Add Connection, AWS, Manage Connections, Connections, Support, and Help Center. At the bottom of the sidebar, it shows the user email 'demo@gmail.com', a 'Logout' link, and a 'Light Mode' toggle. The main content area is titled 'cloudsherpa-staging.gijcs.org/addConnection/aws'. It displays a 'STEP 1 OF 3' wizard titled 'Link your AWS account'. Below the title, it says 'Please log in to your AWS account and create a new IAM user. Enter the information below to begin the process of connecting your account.' There are three input fields: 'Account name' (with 'AWS Connection' as a placeholder), 'Access key ID' (with 'EXAMPLE' as a placeholder), and 'Secret access key' (with a masked placeholder). A blue 'Next' button is at the bottom right of the form.





1. Step 1 - Provide the credentials for IAM user. You can choose what you would like to call the connection (can be changed at any time under Connections).
2. Step 2 - Select the services which you would like CloudSherpa to monitor. The billing in this step is optional and you can also opt in or out of usage monitoring for specific services.
3. Step 3 - Displays the resources that are associated with the services that were in the previous step. You can select which ones you would like CloudSherpa to monitor (can be manager anytime under Connections).

## b) GCP Setup Wizard

The setup wizard consists of 3 steps, which each user has to go through each time they want to establish a GCP connection.

CloudSherpa

Analytics

Dashboard

Add Connection

AWS

GCP

Azure

Manage Connections

Connections

Intelligence

Usage Forecasts

Billing Forecasts

Optimization

Recommendations 2 active

demo@gmail.com

Logout

Light Mode

STEP 1 OF 3

Link your GCP account

Please download the service account key as a JSON file and upload it below to begin connecting your account

Account name

Connection name

Service account key

Drag and drop your file here

or

Click to browse files

Next

CloudSherpa

Analytics

Dashboard

Add Connection

AWS

GCP

Azure

Manage Connections

Connections

Intelligence

Usage Forecasts

Billing Forecasts

Optimization

Recommendations 2 active

demo@gmail.com

Logout

Light Mode

STEP 2 OF 3

Select service

Choose which GCP service you want to monitor.

Opt-in to billing

BILLING EXPORT CONFIGURATION

Account-wide cost scope

Cloud Billing Account ID

Enter the Google Cloud Billing account ID for this connection.

XXXXXXXX-XXXXXX-XXXXXX

Billing Export Dataset ID

Enter the BigQuery dataset that contains the detailed billing export tables.

example\_dataset

SERVICES WE OFFER

Deselect All

GCP/ComputeEngine

PERMISSIONS

The image displays two screenshots of the CloudSherpa web application interface, showing the setup process for monitoring GCP services.

**Top Screenshot: Select service (STEP 2 OF 3)**

The interface shows the 'Select service' step. The user is prompted to 'Choose which GCP service you want to monitor.' There is an 'Opt-in to billing' checkbox. Under 'SERVICES WE OFFER', 'GCP/ComputeEngine' is selected. Under 'PERMISSIONS', the following permissions are listed: Compute Viewer, Service Usage Viewer, Cloud Asset Viewer, and Monitoring Viewer. 'Back' and 'Next' buttons are at the bottom.

**Bottom Screenshot: Select instances (STEP 3 OF 3)**

The interface shows the 'Select instances' step. The user is prompted to 'Select the instance you want CloudSherpa to monitor.' Under 'AVAILABLE RESOURCES', there is a search bar and a table of resources:

| Resource         | Type         | Region           | Tags          | Active/Inactive                     |
|------------------|--------------|------------------|---------------|-------------------------------------|
| cerberus-vpn~... | gce_instance | africa-south1... |               | <input checked="" type="checkbox"/> |
| instance-2026... | gce_instance | africa-south1... | goog-ops-a... | <input checked="" type="checkbox"/> |

Below the table is the 'INGESTION INTERVAL' section, which includes a warning about the recommended interval (3 minutes 20 seconds) and a slider to adjust the interval. The slider is currently set to 3 minutes 20 seconds. 'Back' and 'Finish' buttons are at the bottom.

1. Step 1 - Enter an account name that you can refer to when looking for a particular connection. Along with that upload your service account key (it must be a JSON file).
2. Step 2 - The user has an option of selecting whether they would like to apply billing for this connection or not. They need to select all of the services they would like CloudSherpa to monitor. A list of permissions will and appear that the user would need to grant for your newly created GCP IAM user.
3. Step 3 - Once all of the resources associated with your selected services have been discovered, you can choose which ones you would like to monitor. The ingestion interval slider is dependent on the resources that are chosen. The

interval can be adjusted but if a value selected is below the recommended value, the user will be warned of the consequences.

## c) Azure Setup Wizard

The setup wizard consists of 3 steps, which each user has to go through each time they want to establish an Azure connection.

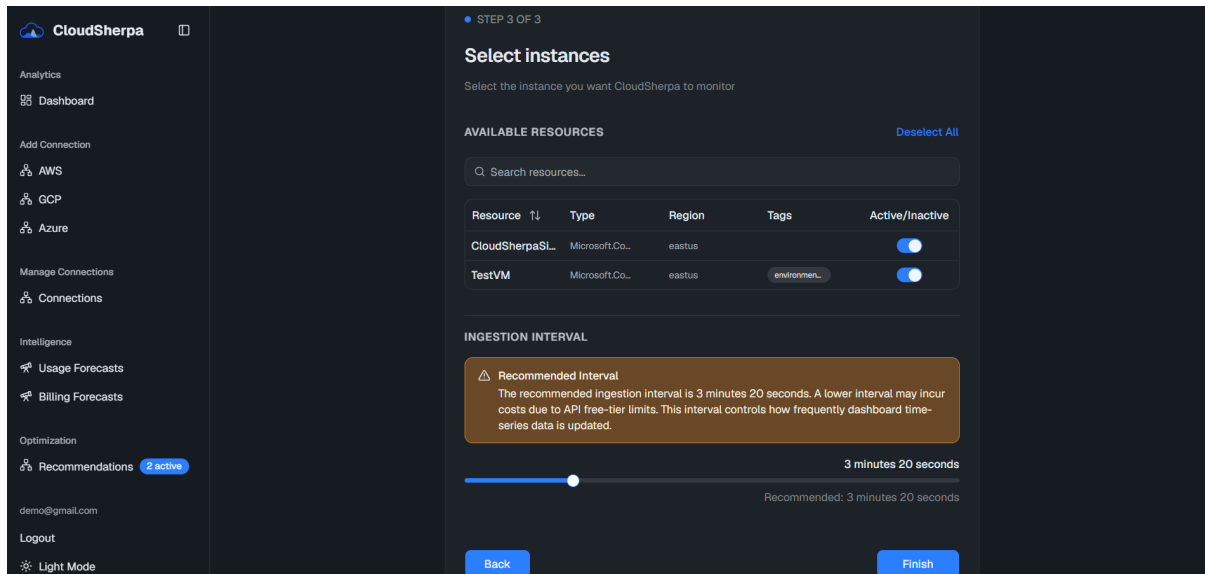
The image displays two screenshots of the CloudSherpa Azure Setup Wizard interface.

**Top Screenshot (Step 1 of 3):**

- Header:** CloudSherpa logo and navigation menu (Analytics, Dashboard, Add Connection, Manage Connections, Intelligence, Optimization, Recommendations (2 active), demo@gmail.com, Logout, Light Mode).
- Step 1 of 3:** Link your Azure account. Follow the steps to begin connecting your account.
- Form Fields:**
  - Account name: Connection name
  - Subscription ID: Azure subscription ID
  - Tenant ID: Tenant ID
  - Client ID: Client ID
  - Client Secret: (password field)
- Buttons:** Next

**Bottom Screenshot (Step 2 of 3):**

- Header:** Same as the top screenshot.
- Step 2 of 3:** Select service. Choose which Azure service you want to monitor.
- Form Fields:**
  - SERVICES WE OFFER: Deselect All, ☒ AZURE/VM
  - PERMISSIONS: - Monitoring Reader, - Reader
- Buttons:** Back, Next

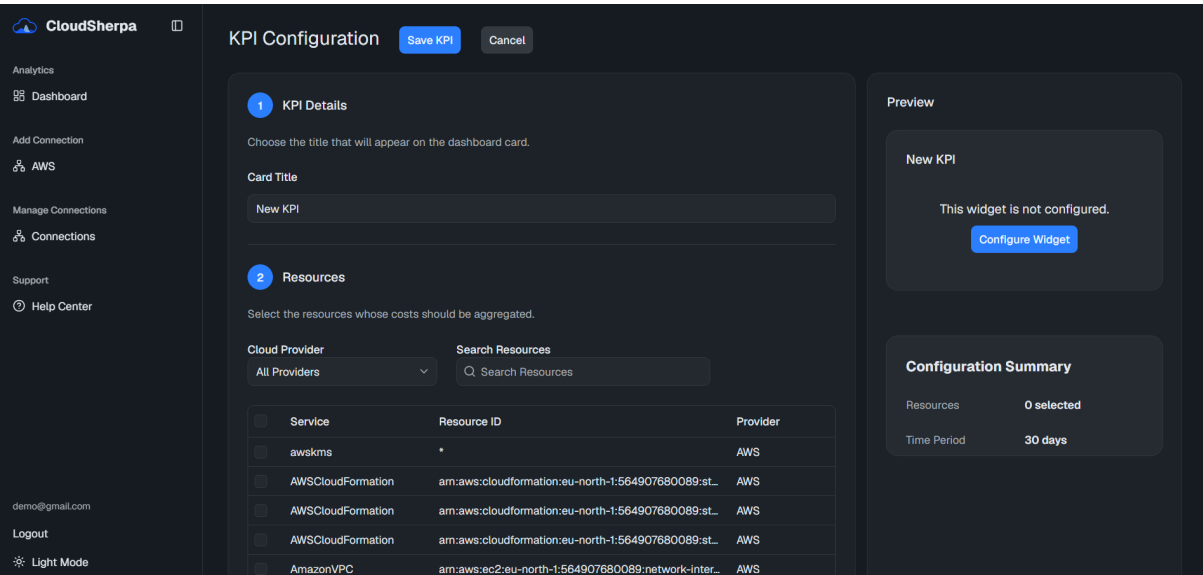
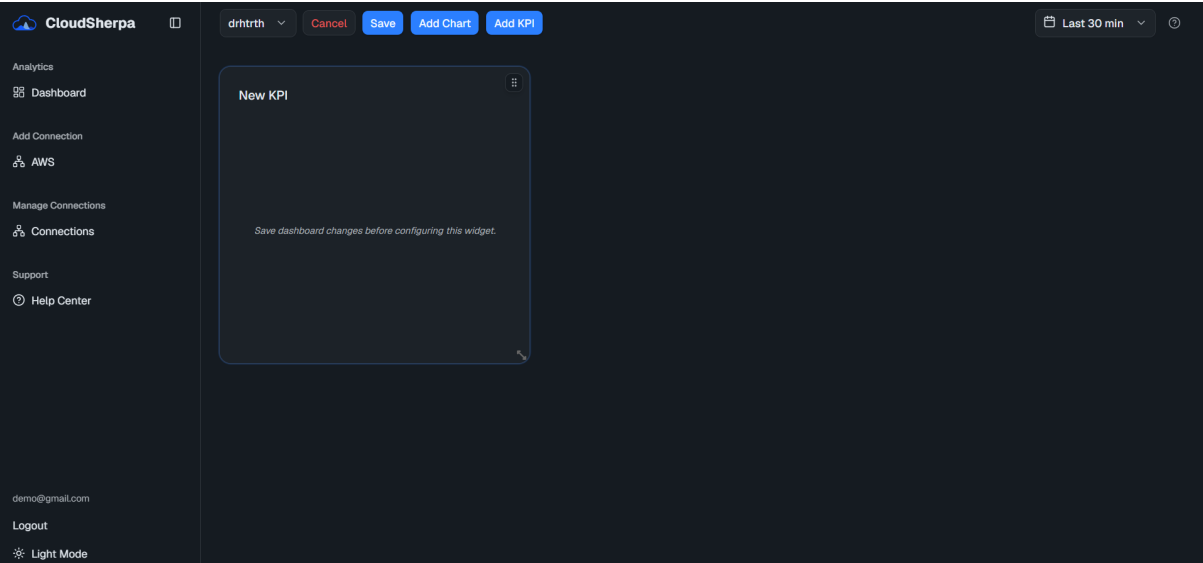


1. Step 1 - Add an account name which you can refer to when looking for a particular connection. Enter your Subscription ID, Tenant ID, Client ID and your Client Secret.
2. Step 2 - Select the services that would be monitored by CloudSherpa and you would then get a list of permissions that you would need to grant.
3. Step 3 - Once all of the resources associated with your selected services have been discovered, you can choose which ones you would like to monitor. The ingestion interval slider is dependent on the resources that are chosen. The interval can be adjusted but if a value selected is below the recommended value, the user will be warned of the consequences.

Some additional features that is available in all of the setup wizards include: Being able to make a certain resource active or inactive for CloudSherpa to monitor. The user can search for resources, Select all of the resources and sort the resources by using the tanstack table feature.

### 3.5 Feature 5: KPI widget, configuration and billing ingestion

It allows the user to view all of their cloud charges that went off against their account. On a granular level they can select which charges to monitor or omit and at a time period that they prefer. This way the user can build a custom view of their cloud spending and gain useful operational insights to make informed decisions.





The screenshot shows the CloudSherpa dashboard with a sidebar on the left containing navigation links: Analytics, Dashboard, Add Connection, AWS, Manage Connections, Connections, Support, and Help Center. The main content area displays a table of AWS resources with columns for Service, Resource ID, and Provider. The table lists various AWS services like AWSKMS, AWS CloudFormation, Amazon VPC, Amazon ECR, AWS Data Transfer, and Amazon ECS. Below the table, there is a pagination control showing '0 of 35 row(s) selected', 'Rows per page: 10', and 'Page 1 of 4'. To the right of the table, there is a 'Time Period' widget set to '30 days'. The sidebar also includes a user profile section with 'demo@gmail.com', a 'Logout' button, and a 'Light Mode' toggle.

| Service            | Resource ID                                           | Provider |
|--------------------|-------------------------------------------------------|----------|
| aws kms            | *                                                     | AWS      |
| AWS CloudFormation | arn:aws:cloudformation:eu-north-1:564907680089:st...  | AWS      |
| AWS CloudFormation | arn:aws:cloudformation:eu-north-1:564907680089:st...  | AWS      |
| AWS CloudFormation | arn:aws:cloudformation:eu-north-1:564907680089:st...  | AWS      |
| Amazon VPC         | arn:aws:ec2:eu-north-1:564907680089:network-inter...  | AWS      |
| Amazon VPC         | arn:aws:ec2:eu-north-1:564907680089:network-inter...  | AWS      |
| Amazon VPC         | arn:aws:ec2:eu-north-1:564907680089:network-inter...  | AWS      |
| Amazon ECR         | arn:aws:ecr:eu-north-1:564907680089:repository/clo... | AWS      |
| AWS Data Transfer  | arn:aws:ecs:eu-north-1:564907680089:task/cloudshe...  | AWS      |
| Amazon ECS         | arn:aws:ecs:eu-north-1:564907680089:task/cloudshe...  | AWS      |

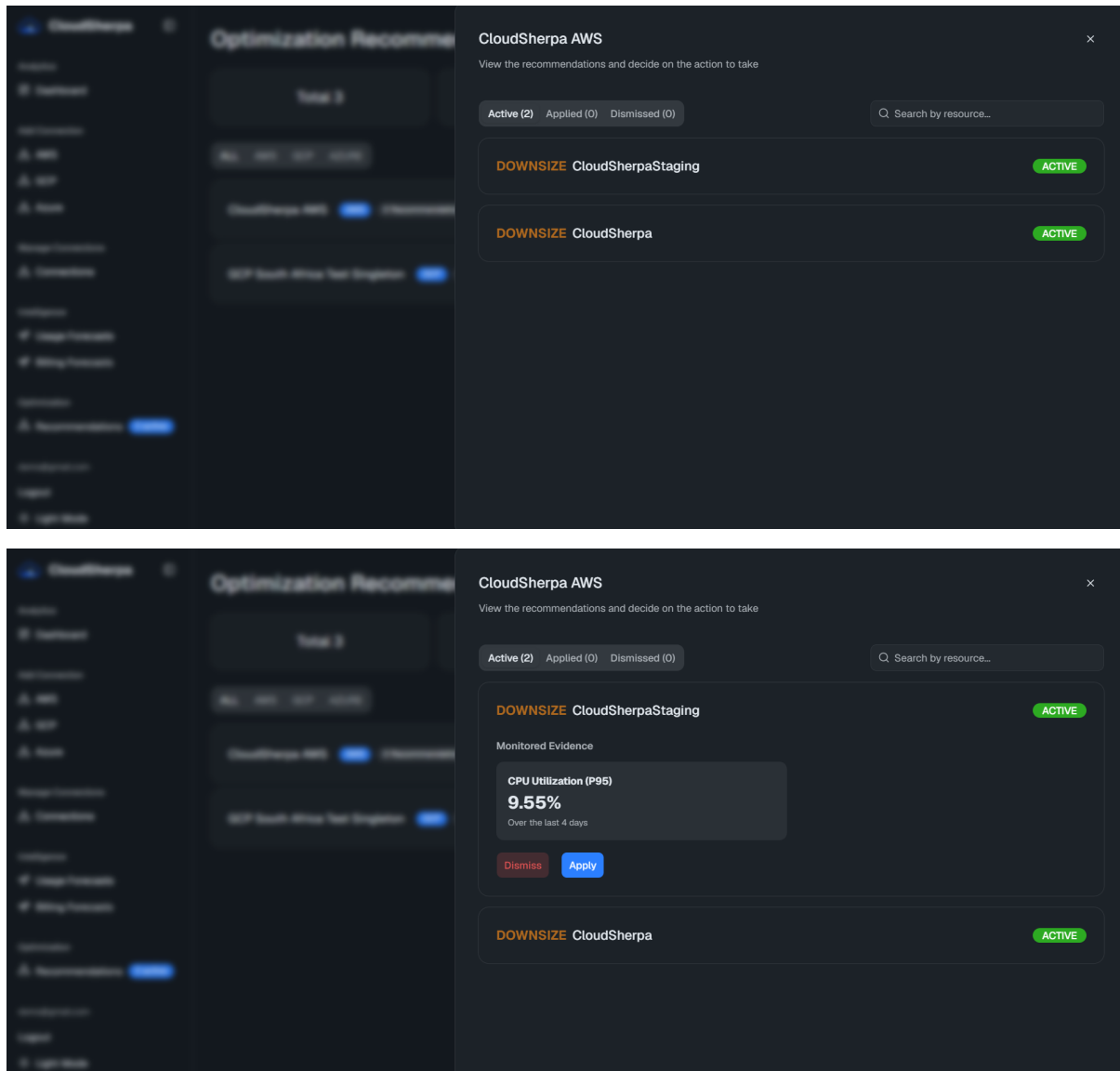
1. The user can change the name of the KPI widget.
2. They can select the specific billing records they would like to monitor.
3. Users can also select which time window they prefer for the billing.

### 3.6 Feature 6: Optimization (Recommendations)

CloudSherpa provides recommendations based on the connections that exist. You can choose whether you would like to accept the recommendation or not.

The screenshot shows the CloudSherpa dashboard with the 'Optimization Recommendations' section active. The sidebar on the left includes navigation links: Analytics, Dashboard, Add Connection, AWS, GCP, Azure, Manage Connections, Connections, Intelligence, Usage Forecasts, Billing Forecasts, Optimization, and Recommendations (2 active). The main content area displays a summary of recommendations: 'Total 3', 'Active 2', 'Applied 0', and 'Dismissed 1'. Below this, there are tabs for 'ALL', 'AWS', 'GCP', and 'AZURE'. A search bar labeled 'Search by Account...' is present. The recommendations are listed in a table with columns for the account name, provider, and the number of recommendations. The first row shows 'CloudSherpa AWS' with '2 Recommendations'. The second row shows 'GCP South Africa Test Singleton' with '1 Recommendation'. The sidebar also includes a user profile section with 'demo@gmail.com', a 'Logout' button, and a 'Light Mode' toggle.

| Account                         | Provider | Recommendations   |
|---------------------------------|----------|-------------------|
| CloudSherpa AWS                 | AWS      | 2 Recommendations |
| GCP South Africa Test Singleton | GCP      | 1 Recommendation  |



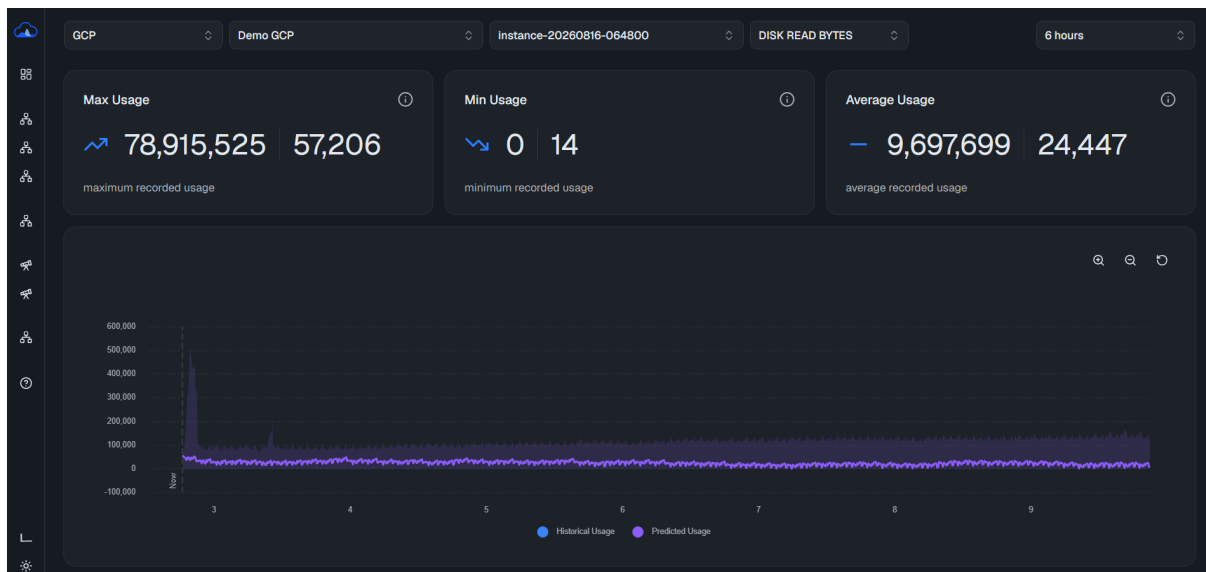
How to navigate to and around the recommendations page:

1. Navigate to the optimizations page using the side navigation bar.
2. A user can click on a connection that is displayed in a list format.
3. A list of recommendations for each resources in that connection will appear.
4. By clicking on a recommendation the user can obtain more information and view the evidence regarding the creation of the recommendation.
5. Click on "Apply" when the user would like to accept that recommendation.
6. Alternatively, the user can click on "Dismiss" to cancel all future recommendation

creations of that type for the specific resource.

## 3.7 Feature 7: Forecasting

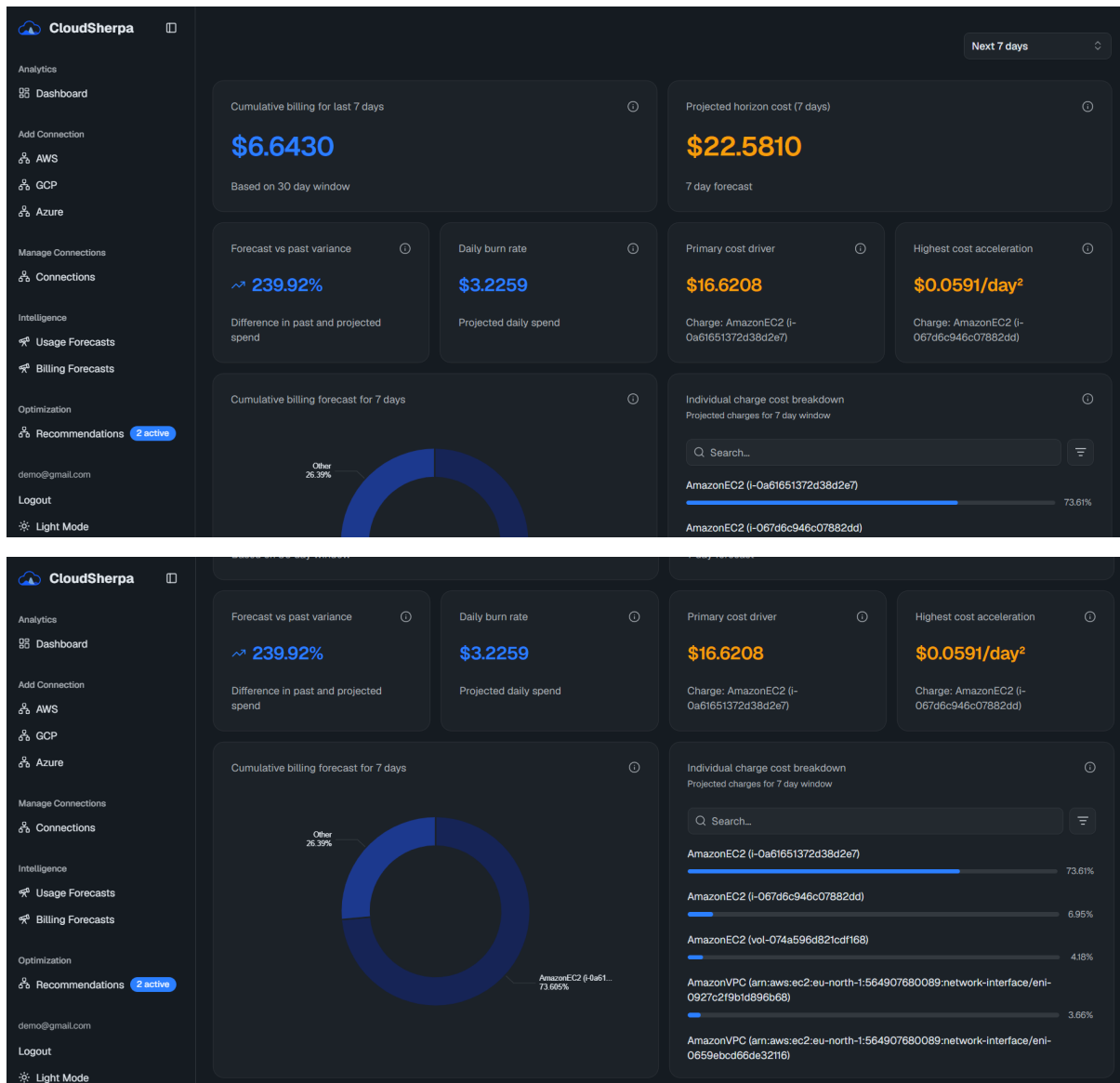
### a) Usage forecasting



How to navigate to and around the Usage Forecasting page:

1. Navigate to the Usage Forecasting page using the side navigation bar.
2. Use the toolbar to select the metric of which resource you would like to view the forecast for.
3. The time period selector importantly only controls historical metrics shown.
4. The cards display the statistics.
5. The shaded area indicates the confidence band and the purple line indicates the forecasted value.

## b) Billing forecasting



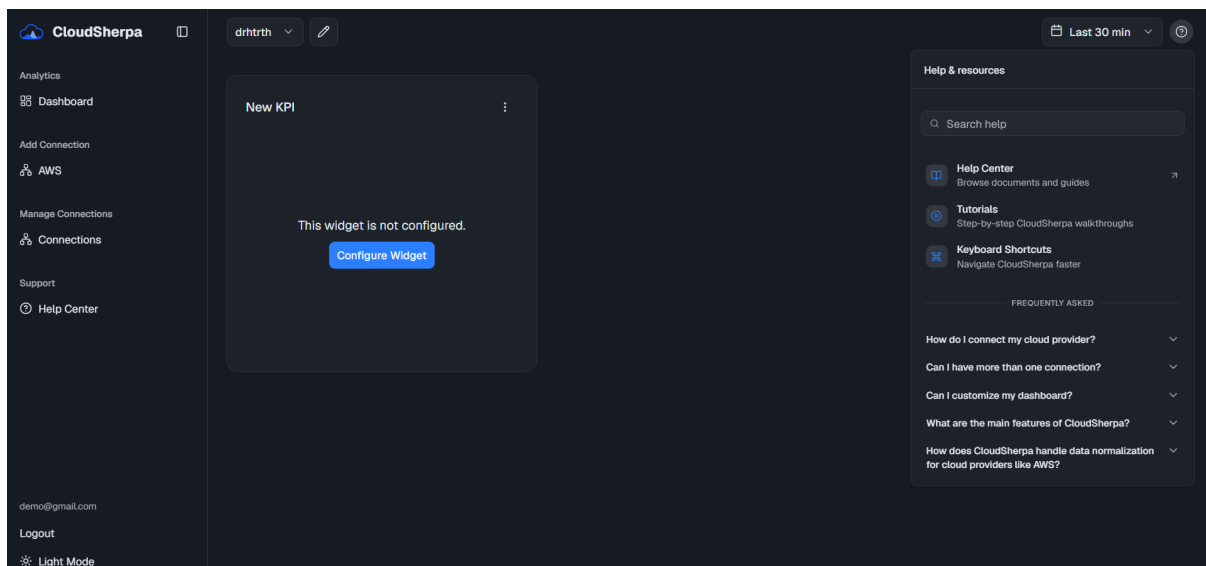
How to navigate to and around the Billing Forecasting page:

1. Navigate to the Billing Forecasting page using the side navigation bar.
2. The user needs to ensure that they have configured billing during setup (this is shown previously in step 2 of the setup wizards).
3. Navigate to billing forecasting page using the side navigation bar.
4. The time period selector, which appears on the top right-hand side corner can be used to control the days that are being forecasted.

## 4 | In-System Help Menu

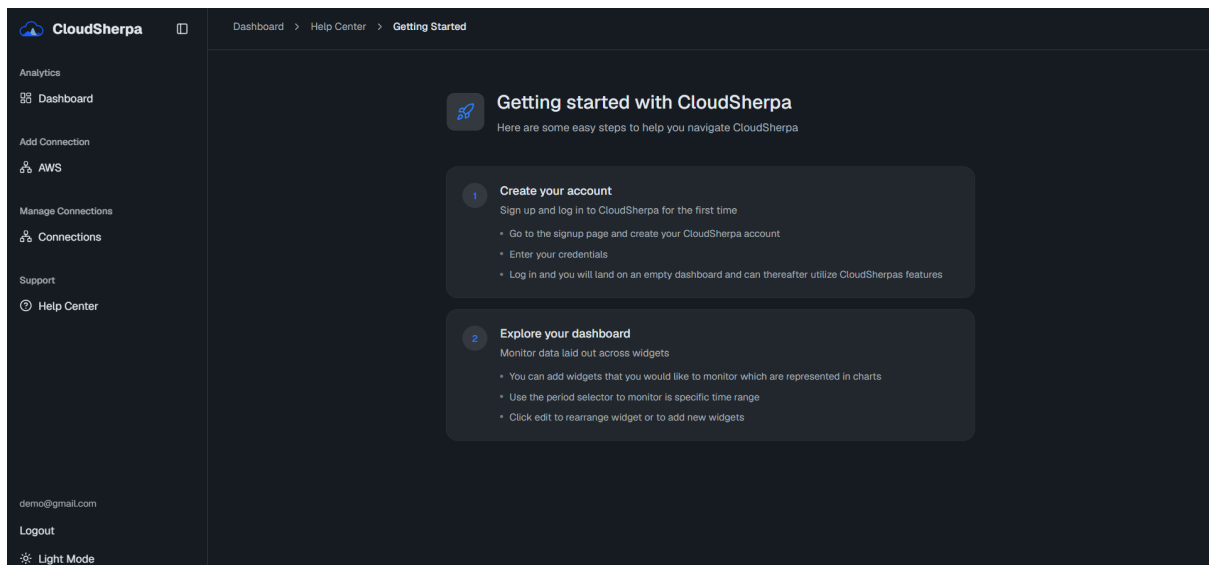
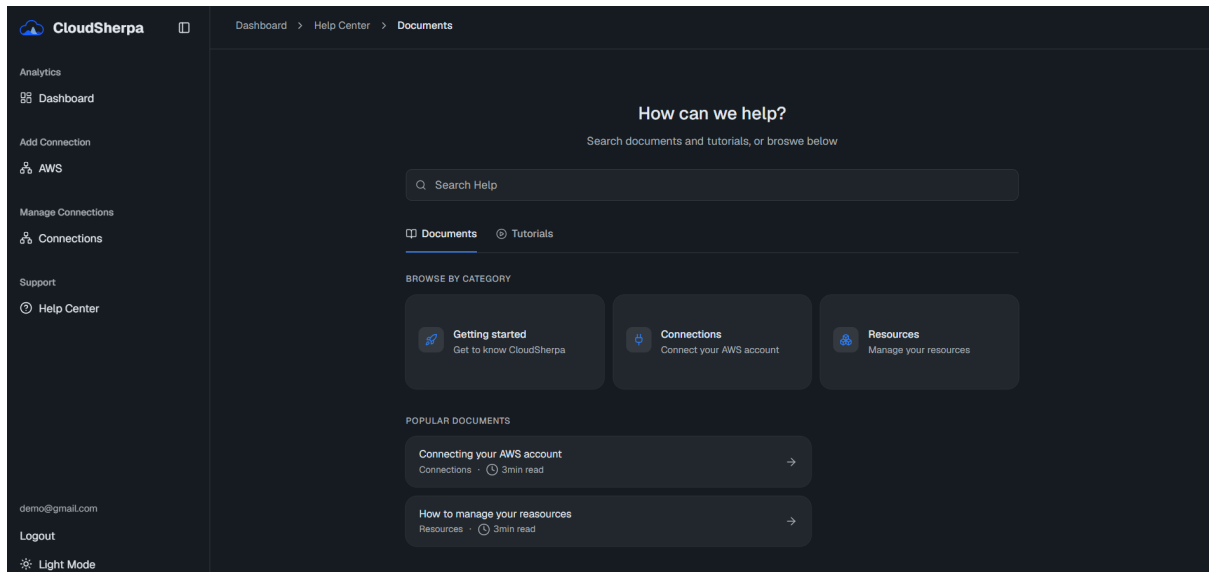
To allow for easy navigation around CloudSherpa, we have included a help menu which users can utilize to familiarize themselves with the features of the platform before delving into it.

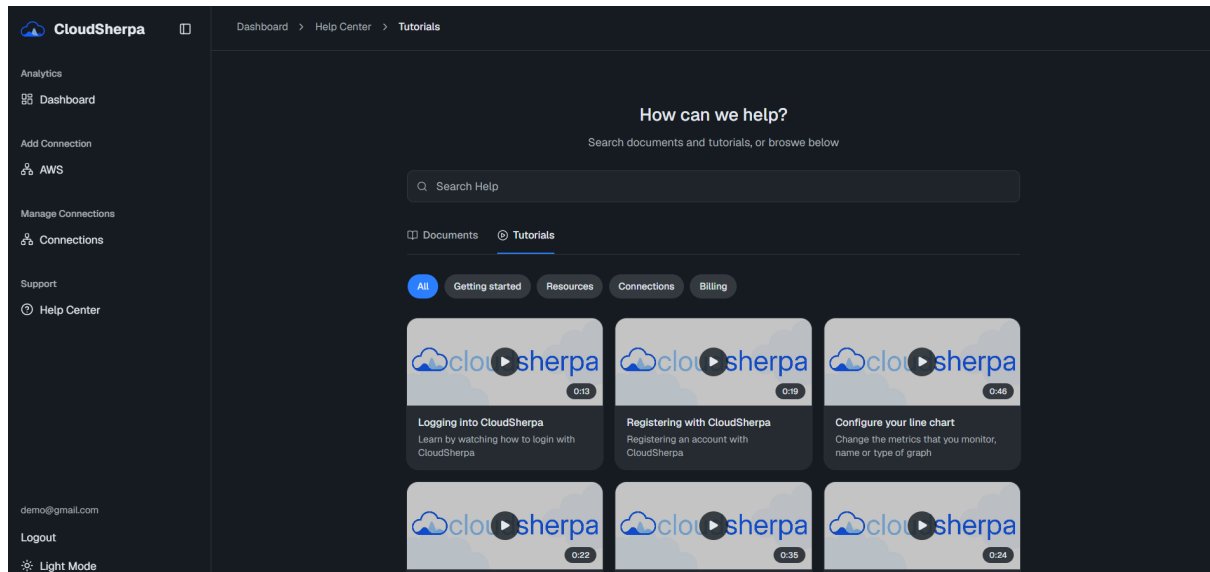
### 4.1 Accessing Support



The help menu can be accessed from the top-right hand corner of the dashboard by clicking on the ? or in the side navigation bar under support.

When clicked upon, the user is provided with a list of options such as: Help center, tutorials, Keyboard shortcuts and frequently asked questions.





Help center provides detailed documentation and tutorial videos which can help a user to understand how the features of CloudSherpa work. The documents provide step by step instructions on how to execute a specific task. It is also accompanied by tutorial videos that demonstrate how the instruction can be executed.